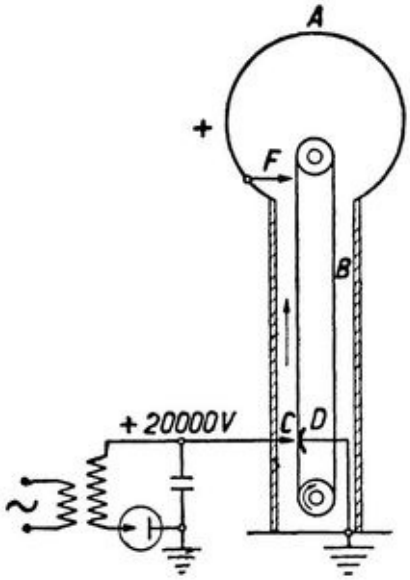


Figure 35.—Van de Graaff's high tension generator.

By far the most efficient apparatus for the production of fast particles is the *cyclotron*, invented by the American Lawrence. It is based on a very interesting principle, that of very frequently repeated acceleration by the same, not very high, voltage; thus it has, among other advantages, the good feature that it dispenses with the high voltages which are so difficult both to obtain and to control. The essential part of the cyclotron is a very large electromagnet which creates a powerful, very homogeneous and wide magnetic field of 10,000–15,000 oersteds between its pole pieces. The pole pieces are placed quite near to each other and the space between them is well evacuated. When a moving particle enters such a field, it describes a circular path, the radius of which is proportional to the velocity of the particle (Figure 37). Therefore, the velocity of the particles is proportional also to the circumference of the circle, and as a result, particles of the same kind, even though having different velocities, require exactly the same time to complete a full revolution. The space between the pole pieces houses two semicylindrical boxes, called *dees*, insulated from each other; between these dees, a potential difference of 30–100 kv is produced by a high-frequency generator. The result is a high-frequency alternating field in the small space between the dees. The frequency of this alternating field is regulated so as to correspond exactly to the period of the revolution of the particles in the magnetic field. The charged particles are made to enter the space between the pole pieces, near the centre (Z). There they come under the influence of the electric field; they attain a certain velocity and move in a semicircle in the space inside a dee where there is a magnetic field only. Proceeding in this manner, they reach the channel between the dees at the exact moment when the electric potential drop there is exactly equal, but opposite in direction, to